Brian Lembuss

Aerospace Engineer | Technical Founder | Deep-Tech Researcher | Academic Educator

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PROFESSIONAL PROFILE

Aerospace Systems Engineer, Technical Founder and Deep-Tech Researcher with six years of progressive experience spanning manned aircraft maintenance, UAV systems engineering, eVTOL flight test management, cloud software architecture, and university instruction. Holds a B.Sc. in Aerospace Engineering with honours from METU and 73 ECTS toward an M.Sc. in Aerospace at TU Munich (degree completion pending). Career includes commercial MRO on turboprop/turbofan aircraft (ALS Ltd); GCS serial production and server-rack HIL test rig builds for commercial UAVs (Amazilia Aerospace); aerodynamic design loop engineering and successful transition flight testing on a 25kg MTOW eVTOL (HORYZN); published airspace throughput research (AIAA 2024); flight simulator instruction (TUM); and founding leadership of Kipepeo Aerospace covering platform CAD, embedded C++ software, GCP cloud microservices, and SAFE equity fundraising. Brings experience across the hardware/software boundary, combining physical aircraft systems lifecycle ownership with cloud, AI, and embedded software architectures.

CORE COMPETENCIES

Aircraft Systems & Aerospace Engineering: Systems engineering lifecycle, Requirements engineering, Functional decomposition, System architecture, Interface definition, Hardware/software integration, Avionics, Flight controls, Embedded systems, HIL testing, Verification & validation, Flight testing, MRO, UAS, eVTOL, Rotorcraft, Fixed-wing aircraft, Aircraft design, Ground control systems, MBSE

PROFESSIONAL EXPERIENCE & SELECTED R&D; PROJECTS

Founding CEO & Lead Systems Engineer — Kipepeo Aerospace (Startup)

Nairobi, Kenya | Oct 2024 – Present

TAI UAS - Tactical Aerial Intelligence Platform (R&D)

- Executed the initial aircraft design lifecycle for the TAI UAS, capturing system-level requirements and conducting preliminary aerodynamic and sizing trade studies for a 2-meter delta-wing hybrid VTOL platform.
- Authored the complete system architecture for the TAI UAS, translating high-level mission requirements into functional allocations, subsystem definitions, and a detailed electrical/control schematic.
- Modeled the 3D mechanical CAD assembly for the TAI delta-wing hybrid VTOL airframe in Autodesk Inventor, incorporating internal structural housing for avionics, propulsion mounts, and modular payload bays.
- Conducted market trade studies, component matching, and cost estimations to compile the full Bill of Materials (BOM) for the TAI UAS platform.

Kilimo Anga - Quadrotor Testbed & AngaCam Payload Integration

- Derived technical specifications and operational requirements for an agile, low-cost quadrotor testbed and multispectral payload based on farmer feedback from the Kilimo Anga initiative.
- Engineered the electrical, power distribution, and flight control architecture for the F450-based quadrotor testbed.
- Designed the 3D mechanical housing and optical mounting for the custom AngaCam multispectral payload in Autodesk Inventor.
- Performed component trade studies and compiled the master Bill of Materials (BOM) for the quadrotor testbed and AngaCam payload.

Kilimo Anga - Field Operations, Pre-Pilot Readiness & Business Model

- Farmer Persona Discovery: Established a validated market intelligence framework and persona matrix that defined Kilimo Anga's target customer segments, feature requirements, and price-sensitivity constraints.
- Competitor Gap Analysis & Fail-Point Mapping: Delivered a comprehensive Competitor Study & Gap Analysis matrix establishing Kilimo Anga's market positioning, pricing advantages, and service-led delivery model.
- Cost of Status Quo (CoSQ) Financial & Impact Modeling: Delivered a defensible financial model and proof-of-impact framework proving value recovery for farmers while establishing unit economics, breakeven thresholds, and investor returns.
- Credit-in-Kind Business Model & Value Chain Offloading Design: Risk-mitigated, market-tailored payment-in-produce mechanism integrated with local grain distribution networks, expanding TAM accessibility without creating unhedged credit risk.

- Pre-Pilot Ecosystem Partnerships, Governance & Investor Readiness: Finalized investment-ready corporate governance baseline, established $30K pre-seed investment positioning, established a 7+ partner execution ecosystem, and captured non-dilutive cloud credit grants from Microsoft and Google.

Linda Nchi - Sovereign Tactical ISR Platform & K-DEMO-2.5 Cargo Demonstrator (R&D)

- Defense Stakeholder Requirement Synthesis & Use-Case Mapping: Delivered a validated stakeholder use-case matrix, regulatory certification pathway, and privacy-compliant data governance framework for sovereign ISR operations.
- Linda Nchi Tactical ISR System Architecture & Technical Roadmap Authoring: Completed a production-ready, 11-section system architecture and technical roadmap establishing subsystem boundaries, interface control targets, and air-gapped on-premise cloud infrastructure schemas.
- Project K-DEMO-2.5 Heavy-Lift Cargo Hexacopter Mechanical Engineering & Stress Analysis: Delivered a fully calculated, production-ready engineering design dossier, CAD geometry layout, vertical CoG stack architecture, and local procurement BOM for the heavy-lift cargo demonstrator.

Aerospace Systems Engineering (Working Student) — Amazilia Aerospace GmbH

Munich, Germany | Sep 2022 – Dec 2024

WfA MiniFreighter GCS -SN2 (R&D)

- Attended joint design evaluation session with Wings for Aid CEO and chief pilot alongside Amazilia CTO and senior systems engineer. Captured customer and pilot requirements directly and translated them into formal engineering specifications for the SN2 GCS redesign. Produced wiring diagrams, CAD renders and system architecture documentation.
- Sourced alternative panel supplier through independent research, which reduced component costs. Redesigned front panels in Autodesk Inventor. Performed mechanical assembly including drilling connection holes on panels and GCS case, installing internal support structures, and routing the power architecture.
- Performed stepwise electrical integration of all internal GCS components following the assembly procedure, conducting QA checks and functionality tests. Scheduled and conducted simulator testing in the lab to evaluate control software, internal wiring of controls, stick feedback and aircraft response time.
- Serially produced four additional GCS units following customer acceptance of Unit 1. Delivered 5 units total to Wings for Aid for operational deployment with WFP in Africa.

Amazilia Ground Control Station (R&D)

- Reviewed and cleaned up existing CAD files, wiring diagrams and system architecture documentation inherited from predecessor.
- Performed mechanical assembly of three tabletop GCS units including mounting of three-monitor arrangement, mode control panel and internal structural components.
- Performed electrical integration of all internal components across three units including power architecture, monitor connections, mode control panel wiring and communication interfaces.
- Conducted functional testing of each completed unit, verifying correct operation of all three monitor feeds - camera, telemetry and engine parameters - across Pipistrel OPV, Nuuva and CTOL platforms.

Aircraft Systems Hardware-in-the-Loop (HIL) Test Rig (R&D)

- Received system requirements defining the HIL rig scope. Developed the high-level wiring architecture covering OPS Panel and PSUs, CAN boxes, RPi and USB hub, FCCs and DCUs, ETH switch, LMUs, Panels A/B/C and BRS tray. Produced detailed wiring sheets and interface definitions across all components.
- Designed the full server rack assembly in Autodesk Inventor, modelling the physical layout and component placement of the HIL rig.
- Procured all components for the HIL rig. Performed mechanical assembly of the server rack.
- Performed full electrical integration of all HIL rig components including power distribution, CAN bus wiring, Ethernet patch panel connections, and interface wiring between flight hardware.
- Conducted extensive testing and verification of the completed HIL rig, validating correct operation of all interfaces, communication protocols and hardware components simulating aircraft systems.

Aircraft Battery Charging Unit (ABCU) (R&D)

- Following a polarity reversal incident during field testing of a Pipistrel aircraft, conducted requirements capture to design a safe standalone aircraft battery charging unit (12V and 24V LiFePO4).
- Developed system architecture for the ABCU covering power input, charging system selection, voltage switching between 12V and 24V configurations, polarity protection, and battery voltage indication.
- Produced full wiring diagrams for the ABCU internal architecture. Designed enclosure panels and overall box layout in Autodesk Inventor.
- Compiled and handed over the complete ABCU design package (architecture diagrams, wiring diagrams, component datasheets, BOM, panel CAD designs, and assembly guide) to the team upon departure.

Project Manager - Systems Integration & Flight Testing — HORYZN (TU Munich Student Aerospace Initiative)

Munich, Germany | Jul 2023 – Jul 2024

Kolibri eVTOL — Systems Integration, Flight Testing & Project Completion

- Appointed Project Manager for systems integration and flight testing phase of Kolibri hybrid lift-and-cruise eVTOL. Obtained EU A1/A3 drone pilot licence.
- Served as the central coordination point between avionics, flight testing, CAD and structures teams. Identified and resolved physical integration conflicts.
- Worked hands-on with the avionics and flight testing team on the integration of Kolibri's avionics systems including flight controller setup on PX4, ESC configuration, sensor integration and onboard electronics wiring.
- Participated in ground testing and hover trial campaign for Kolibri. Built and operated a hover test bench to validate lift motor thrust ratios against design requirements for the 25kg MTOW aircraft. Tested the defibrillator payload delivery system. Analysed flight logs to drive corrective actions.
- Supported preparation of regulatory documentation for Kolibri flight operations under the SORA framework.
- Secured access to a TU Munich aerospace test facility (Oberpfaffenhofen) for the Kolibri flight test campaign, managing logistics, scheduling, and sponsor communications.
- Participated in the final Kolibri transition flight test campaign at Oberpfaffenhofen test airport, achieving successful VTOL hover, transition to forward flight, and cruise phases-concluding Mission Pulse Phase 2.

Aerodynamics Project Engineer — HORYZN (TU Munich Student Aerospace Initiative)

Munich, Germany | Oct 2022 – Jun 2023

Kolibri eVTOL — Aerodynamics Module Development

- Participated in structured handover sessions with the outgoing Mission Pulse Phase 1 team to understand the design, shortcomings and failure modes of the Frankenstein prototype.
- Led the aerodynamics team's contribution to the Kolibri configuration selection process. Identified the root cause of Frankenstein's transition failure and proposed a Cessna-type fixed-wing configuration with a front tractor propeller and lift motors mounted on the wing, successfully securing adoption.
- Developed and maintained the aerodynamics module within the HORYZN multidisciplinary design loop, interfacing with CPACS via TiGL/TiXi libraries, using PAWAT within a MATLAB class-based architecture.
- Participated in airfoil selection (XFLR5/Flow5) and ran iterative optimization of wing geometry (twist, span, chord, sweep).
- Worked intensively within the RCE-based multidisciplinary design loop to achieve design convergence for Kolibri.
- Prepared and delivered a comprehensive handover of the aerodynamics module to the incoming team (codebase, UML, I/O specifications).

UAS Engineer, Pilot and Instructor — Kendrone Ltd

Mtwapa, Nairobi, Naivasha | Jan 2021 – Jun 2021

Drone-Based Seedball Dispersal Mechanism (R&D)

- Captured mission requirements from CEO and Seedball Kenya and produced mechanical design of seedball container and servo-actuated release mechanism in Autodesk Inventor.
- Fabricated mechanism components (3D printing) and integrated the dispersal payload onto the Tarot 650 airframe including servo power rail decoupling.
- Prepared Tarot 650 platform for flight operations, resolving flight controller calibration/imbalance issues and configuring PX4 failsafes.
- Executed 3-flight endurance test campaign proving optimal dual 6S parallel battery setup (~9 min endurance, ~2 km grid coverage per sortie). Authored formal test report.
- Conducted a witnessed operational demonstration for Seedball Kenya and observers.

Aerial Mapping & Agricultural Survey Operations

- Executed end-to-end aerial survey and agricultural mapping missions as PIC (DJI Phantom 4 Pro RGB/Multispectral), processing data in Pix4D/Pix4Dfields to produce orthomosaics, DSM/DTM, and vegetation index outputs.
- Trained professional surveyors from external survey companies in UAS-based mapping workflows during live operational missions.

UAS Avionics, Payload Integration & Fleet Maintenance

- Integrated avionics and sensor payloads across the Kendrone multirotor fleet.
- Diagnosed and performed component-level hardware repairs (IMU, gimbals, motor/ESCs) on DJI platforms to restore airworthiness.

- Conducted systematic pre- and post-flight airworthiness inspection routines across the Kendrone fleet with zero platform-related mission failures.

Aircraft Maintenance Intern — ALS Ltd

Nairobi, Kenya | Jul 2017 – Sep 2017

Aircraft Line, Base & C-Check Maintenance - ALS Ltd MRO

- Assisted with fuel nozzle replacement, compressor wash, and propeller shaft greasing on the PT6A-65B turboprop engine (Beechcraft 1900C 200HR inspection).
- Assisted with de-icer boot reinstallation and flap ball screw assembly lubrication on the DHC-8 during line maintenance.
- Assisted with cabin seat removal, hook-and-loop fastener replacement, and nose landing gear servicing on the Embraer ERJ145MP during C-check.

ADDITIONAL RESEARCH & ACADEMIC PROJECTS

Generic Modeling of Slotneutral UAM Throughput at Commercial Airports (R&D) — Technical University of Munich

2023 – 2024

- Processed 24-hour Flightradar24 traffic data (Munich Airport) showing 86.9% of arriving traffic falls into CAT-D/E. Formulated IFR PinS approach procedures allowing eVTOLs to transition without crossing active runways. Delivered a Python toolchain generating 3D KML trajectories, reducing approach detours from 66 km to ~15 km. Co-authored the resulting paper presented at AIAA 2024.

Autonomous Sub-Terrain UAV Challenge (R&D) — Technical University of Munich

2023 – 2024

- Engineered a depth-to-pointcloud perception engine and an OctoMap 3D voxel grid mapping pipeline in ROS/C++ for a simulated sub-terrain quadrotor executing fully autonomous cave transit and object localization.

Embedded Software Development of the Actuator Control and Monitoring Unit (ACMU) (R&D) — Technical University of Munich

2024

- Refactored the embedded C++ data logging software pipeline for an aircraft ACMU using TDD. Implemented high-efficiency binary serialization over USB, achieving an 80% reduction in frame size (100 to 20 bytes).

Conceptual Design of a Fixed-Wing Luxury Aircraft (Capstone I) — Middle East Technical University

2018 – 2019

- Delivered a fully sized conceptual design (W0 = 20,600 kg) validated against performance constraints, with 1:150 scale 3-view CAD layouts.

Conceptual Design of a VTOL Personal Air Vehicle (Capstone II) — Middle East Technical University

2018 – 2019

- Sized a light civil rotorcraft (TOGW = 2,014 kg) using BEMT, completing forward flight power analysis and generating a full configuration design.

Design and Development of an Automated Air Filter — Middle East Technical University

2019

- Designed, built, and tested an autonomous Arduino-based air filtration system detecting dust/smoke density.

MBSE Agricultural UAS Architecture (R&D) — University of Cologne / DAAD

2022 – 2023

- Developed a comprehensive MBSE-structured UAS architecture for precision agriculture at the Galana Kulalu Irrigation Scheme, complete with payload definitions, WBS, and SWOT/PEST analyses. Graded Sehr gut (A).

EDUCATION

M.Sc. Aerospace Engineering — Technical University of Munich (TUM) (Munich, Germany)

Oct 2021 – Present

Status: Degree completion pending (73 ECTS completed; 17 credits and thesis remaining).

B.Sc. Aerospace Engineering — Middle East Technical University (METU) (Northern Cyprus Campus)

Sep 2015 – Jun 2019

Status: Graduation Honors: ■eref / Honour Classification (CGPA: 3.12/4.00, 259.5 ECTS).

RESEARCH & PUBLICATIONS

- **Co-Author**, *"Conceptualization and simulation of Advanced Air Mobility (AAM) operations within Controlled Airspaces of airports running independent parallel runway operations"* — AIAA 2024 Aviation Forum

LEADERSHIP & MENTORSHIP

Vice President — Aerospace Society, METU

Sep 2017 – Jun 2018

Co-managed administrative operations, launched the model aircraft project team, and facilitated MATLAB/Inventor technical courses.

LICENCES & CERTIFICATIONS

- Private Pilot Licence - Aeroplane (YK-9469-PL) | KCAA / Skylink Flight Services (2015)
- Flight Radio Telephony Operator's Licence (YK-9469-RL) | KCAA (2015)
- English Language Proficiency Rating (Level 5) | KCAA (2015)
- Remote Pilot Licence - Multirotor (YK-RPL-00013A) | KCAA / Kendrone Ltd (2021)
- Instructor Rating | KCAA / Kendrone Ltd (2021)
- EU A1/A3 Drone Pilot Licence | EASA (2023)
- Full Stack Software Development Certificate | eMobilis Technology Institute (2020)

LANGUAGES

English: Native / Fluent · **Swahili:** Native / Fluent · **German:** B2